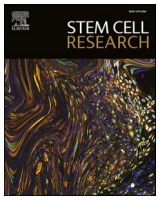
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Lab Resource: Multiple Cell Lines



Generation of three isogenic, gene-edited iPSC lines carrying the APOE-Christchurch mutation into the three common APOE variants: *APOE2Ch*, *APOE3Ch* and *APOE4Ch*

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A B S T R A C T

Late-onset Alzheimer's disease (AD) has become the paradigm of a non-mendelian complex neurodegenerative disease, for which a major genetic determinant is known, the APOE locus. A rare APOE variant named *Christchurch* (*APOEch*) yielding a missense mutation from Arginine to Serine at amino acid 136, has been suggested to exert a protective effect in an individual carrying the most penetrant form of Familial AD (Paisa mutation in PSEN1 gene, E280A). We describe here a new set of induced pluripotent stem cell (iPSC) lines, where the Christchurch mutation (Ch) has been introduced by gene editing into the APOE locus of three isogenic iPSC lines carrying the more common APOE variants (APOE 2/2, APOE 3/3, and an APOE 4/4) in homozygosity. Brain cells derived from these iPSC lines will enable a better understanding of APOE biology in general and facilitate the study of how the *Christchurch* variant affects the function of each APOE genotype. This set of iPSC lines are globally available via the European Bank of iPSCs, EBiSC.org.

1. Resource table

Unique stem cell lines identifier	BIONi010-C-70 The line has an APOE 2/2 genotype with an additional, homozygous Christchurch mutation BIONi010-C-71 The line has an APOE 3/3 genotype with an additional, homozygous Christchurch mutation BIONi010-C-72 The line has an APOE 4/4 genotype with an additional, homozygous Christchurch mutation
Alternative name(s) of stem cell lines	1) APOE2Ch 2) APOE3Ch 3) APOE4Ch
Institution	Bioneer-Janssen Pharmaceutica
Contact information of distributor	Benjamin Schmid-Alfredo Cabrera-Socorro
Type of cell lines	iPSC
Origin	Human
Additional origin info required for human ESC or iPSC	Age: 18 Sex: Male Ethnicity if known: African
Cell Source	iPSC

(*continued on next column*)

(*continued*)

Unique stem cell lines identifier	BIONi010-C-70 The line has an APOE 2/2 genotype with an additional, homozygous Christchurch mutation BIONi010-C-71 The line has an APOE 3/3 genotype with an additional, homozygous Christchurch mutation BIONi010-C-72 The line has an APOE 4/4 genotype with an additional, homozygous Christchurch mutation
Clonality	Clonal
Method of reprogramming	CRISPR
Genetic Modification	YES
Type of Genetic Modification	Single base change
Evidence of the reprogramming transgene loss (including genomic copy if applicable)	Sanger Sequencing, SNP Array
Associated disease	Alzheimer's Disease
Gene/locus	APOE; 19q13.32; rs7412; rs429358; rs121918393
Date archived/stock date	02/11/2022
Cell line repository/bank	https://www.ebisc.org/
Ethical approval	https://www.ebisc.org/

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2. Resource utility

New iPSC-based tools should facilitate the study of how APOE variants impact AD and other human conditions. We describe a new set of iPSC lines homozygous for the Christchurch mutation that was edited within each APOE known variant. These lines are accessible to the scientific community via the public repository.

3. Resource details

Amongst the few known risk factors for AD, Apolipoprotein E (APOE) genotype stands as the strongest genetic risk of late-onset AD (LOAD), which accounts for up to 95 % of all AD cases. APOE protein has three major isoforms (APOE2, APOE3 and APOE4) depending on the combination of two polymorphisms located at positions 112 and 158 of the APOE-protein, respectively. Having a single APOE4 allele increases the risk 2- to 4-fold and having two APOE4 alleles increases the risk about 8- to 12-fold, although risk varies according to genetic background and sex (Rajabli et al., 2022). On the contrary, APOE2 genotype is protective for AD (Serrano-Pozo et al., 2021). Recently, another APOE variant known as Christchurch has been suggested to be protective for the most penetrant form of Familial AD caused by a mutation in PSEN1 gene (Arboleda-Velasquez et al., 2019). Although its best characterized function is to be a cholesterol transporter, APOE biology is not well understood. Its role in the central nervous system (CNS) may differ from the one in blood as the protein is locally produced in both the central and peripheral compartments and does not cross the blood brain barrier. Induced pluripotent stem cells (iPSCs) facilitate the study brain cell biology *in vitro* as these can be efficiently differentiated into neurons and glia. In this report, we detail the gene editing of three independent, isogenic iPSC lines carrying the three most known APOE variants in homozygosity to introduce the Christchurch variant also in homozygosity. The so-called Christchurch mutation in the *APOE* gene (corresponding to R136S amino acid change) was inserted along with an *APOE* 2/2, *APOE* 3/3 and an *APOE* 4/4 genotype in both alleles. In the first part of the project, a part of the APOE gene, encoding both the site for the Christchurch SNP as well as the APOE SNPs, was removed. This line was used as a master cell line, in which the missing part was edited to introduce the sequences of interest. One clone for each APOE genotype was found and banked (Table 1).

After gene-editing, the identity of all clones was confirmed by STR analysis (submitted to Journal, data not published). One day after thawing, survival, and typical iPSC morphology of the three clones were confirmed by brightfield microscopy (Fig. 1A). All clones showed a normal karyotype by G-banding analysis, and no aberrations were found by SNP array analysis (Fig. 1B). Pluripotency was assessed by flow cytometry analysis for the stem cell markers OCT4, SOX2, SSEA4, and TRA 1–81 at the pre-differentiated stage with the cells being negative for the differentiation marker SSEA1 (Fig. 1C). On each one of the three clones, we also assessed differentiation potential by direct differentiation to all three germ layers followed by flow cytometry analysis of markers defining each germ layer: SOX17 for endoderm, CD56 for mesoderm and PAX6 for ectodermal lineage (Fig. 1D). The genotype of each clone was further confirmed by sanger sequencing (Fig. 1E), and absence of off-target effects above 500 bp were assessed by optical genome mapping using Saphyr® System (Bionano Genomics, supplementary figure).

Table 1
Characterization and validation.

Classification	Test	Result	Data
Morphology	Photography Bright field	Normal	Fig. 1 panel A
Phenotype	Qualitative analysis	Flow cytometry staining of pluripotency markers: Oct4, SSEA-4, SSEA-1	Fig. 1 panel B
Genotype	Karyotype (G-banding) and resolution	46XY, Resolution 400-500	Fig. 1 panel B
Identity	STR analysis	16 sites tested. All matched	Supplementary file 2
Mutation analysis (IF APPLICABLE)	Sanger Sequencing	Homozygous insertion of the Christchurch mutation (rs121918393)	Fig. 1 panel E
	SNP Array	No off-target insertions	Fig. 1 panel B
Microbiology and virology	Mycoplasma	Mycoplasma testing by PCR	Supplementary file 1
Differentiation potential	Directed differentiation and tri-lineage differentiation test	Flow cytometry staining of differentiation markers: PAX6 (Ectoderm), CD56 (Mesoderm), and SOX17 (Endoderm)	Fig. 1 panel D

4. Materials and methods

4.1. Design and ordering of tools

The iPSC line BIONi010-C was used for the generation of the three gene-edited clones (Schmid et al., 2019). Two sgRNAs were designed targeting a part of exon4 of the APOE gene. This line was then used as a master cell line. SNP analysis confirmed that no major aberration occurred in this line, and the line was used for further gene-editing. A new sgRNA was designed recognizing the sequence of the master cell line with the deletion in exon4. ssODNs were used as homologous constructs with the genotypes APOE2Ch, APOE3Ch or APOE4Ch (Table 2).

4.2. Gene-editing

In the first part, a small piece of the APOE gene was cut out using a combination of the two CRISPRs "Cr3_APOE_112-158-del_FW" and "Cr4_APOE_112-158-del_RV". Successful deletion was confirmed by PCR using the primers "APOE_E4_FW" and "APOE_E4_RV". The PCR product was sequenced using the sequencing primer "Seq112 FW primer". After confirming the deletion by sequencing, the clone was sent to SNP analysis. No aberrations were detected, and the clone was used for the second part of gene-editing. In the second part, the missing piece of the APOE gene was replaced with the desired APOE genotype and the Ch mutation. The master cell line was therefore nucleofected three times in separate experiments with a combination of the CRISPR "Cr5_APOE_rec-del_RV" and either "ssODN_APOE22_CC_250_FW", "ssODN_APOE33_CC_250_FW" or "ssODN_APOE44_CC_250_FW" as homologous template. Reinsertion of the initially deleted part was confirmed by PCR and sequencing using the same primers as in the first part. For each genotype, one clone was found and banked.

4.3. Cell banking

iPSCs were grown in three 15 cm plates on matrigel in E8 medium to a density of 80 %. Cells were detached with 0.1 % EDTA and centrifuged at 120g for 5 min. The cell pellet was resuspended in 50 mL of freezing medium (50 % E8, 40 % FCS, 10 % DMSO), and 1 mL aliquots were distributed in cryo vials. The vials were transferred in isopropanol containers into a − 80 °C freezer over-night. For long term storage, the cells were transferred into a nitrogen tank.

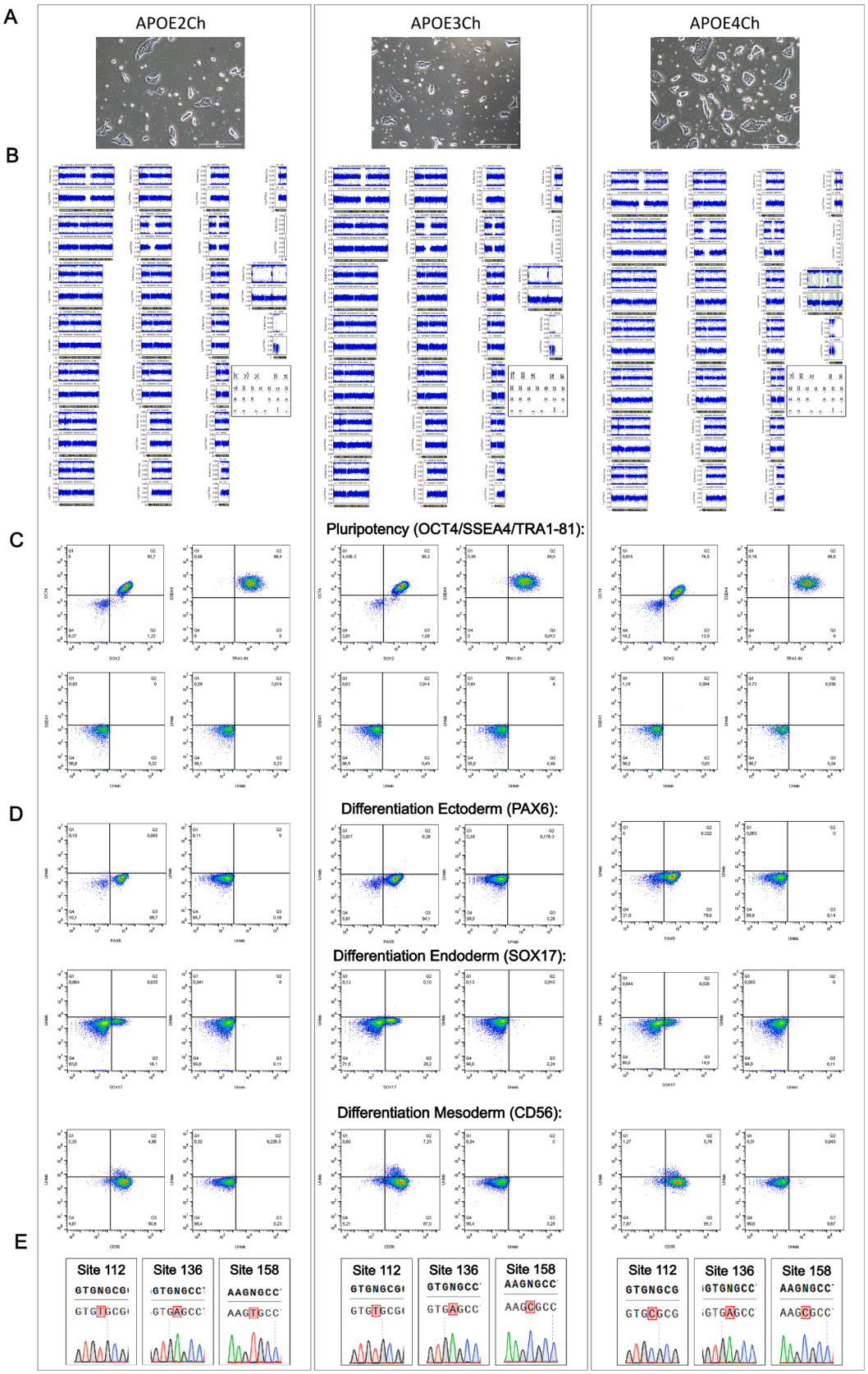

A
APOE2Ch
APOE3Ch
APOE4Ch
B
C
Pluripotency (OCT4/SSEA4/TRA1-81):
D
Differentiation Ectoderm (PAX6):
Differentiation Endoderm (SOX17):
Differentiation Mesoderm (CD56):
E
Site 112
GTGNGCG
GTGTGCG
GTGCGCG
Site 136
GTGNGCC
GTGAGCC
Site 158
AAGNGCC
AAGTGCC
AAGCGCC

Fig. 1.

Table 2
Reagents details.

	Antibody	**Company Cat #**	**RRID**
Pluripotency Marker	OCT4 PE	BD Pharmigen Cat:560,186	RRID:AB_1645331
Pluripotency Marker	Mouse anti-TRA-1-81	Biolegend Cat: 330702	RRID: AB_1089240
Pluripotency Marker	Mouse anti-SSEA4	Biolegend Cat:330,402	RRID:AB_1089208
Differentiation Marker	SSEA1 PE	BD Pharmigen Cat: 560,142	RRID:AB_1645246
Differentiation Marker	CD56 APC	BD Pharmigen Cat:555,518	RRID:AB_398601
Differentiation Marker	SOX17 AF647	BD Pharmigen Cat:562,594	RRID:AB_2645326
Differentiation Marker	PAX6 AF647	BD Pharmigen Cat:562,249	RRID:AB_11152956
Secondary antibodies	Donkey Anti-Mouse IgG Alexa fluor 488	Life technologies Cat#A21202	RRID: AB_141607
Secondary antibodies	Donkey Anti-Mouse IgG Alexa fluor 647	Invitrogen Cat#A31571	RRID:AB_162542
	Target	**Forward/Reverse primer (5′-3′)**	
Primer and CRISPR sequences	APOE_E4	CACTGTGCGACACCCTCC/ GCTTCGGCGTTCAGTGATTG	
Primer and CRISPR sequences	Seq primer Seq112	ACAAATCGGAACTGGAGG	
Sequences for sgRNAs (CRISPRs)	Cr3_APOE_112-158-del_FW	GGCGCAGGCCCGGCTGGGCGCGG	
Sequences for sgRNAs (CRISPRs)	Cr4_APOE_112-158-del_RV	CCCCGGCCTGGTACACTGCCAGG	
Sequences for sgRNAs (CRISPRs)	Cr5_APOE_rec-del_RV	CCGGCCTGGTACACTCCAGCCGG	
Sequences for ssODNs	ssODN_APOE22_CC_250_FW	CACGGCTGTCCAAGGAGCTGCAGGCGGCGCAGGCCCGGCTGGGCGCG-GACATGGAGGACGT-GTGCGGCCGCCTGGTGCAGTACCGCGGCGAGGTGCAGGCCATGCTCGGCCAGAGCACCGAGGAGCTGCGGGTGAGCCTCGCCTCCCACCTGCGCAAGCTGCG-TAAG-CGGCTCCTCCGCGATGCCGATGACCTGCAGAAGTGCCTGGCAGTGTACCAGGCCGGGGCCCGCGAGGGCGCCGAGCGCGGCCT
Sequences for ssODNs	ssODN_APOE33_CC_250_FW	CACGGCTGTCCAAGGAGCTGCAGGCGGCGCAGGCCCGGCTGGGCGCG-GACATGGAGGACGT-GTGCGGCCGCCTGGTGCAGTACCGCGGCGAGGTGCAGGCCATGCTCGGCCAGAGCACCGAGGAGCTGCGGGTGAGCCTCGCCTCCCACCTGCGCAAGCTGCG-TAAG-CGGCTCCTCCGCGATGCCGATGACCTGCAGAAGCGCCTGGCAGTGTACCAGGCCGGGGCCCGCGAGGGCGCCGAGCGCGGCCT
Sequences for ssODNs	ssODN_APOE44_CC_250_FW	CACGGCTGTCCAAGGAGCTGCAGGCGGCGCAGGCCCGGCTGGGCGCG-GACATGGAGGACGT-GCGCGGCCGCCTGGTGCAGTACCGCGGCGAGGTGCAGGCCATGCTCGGCCAGAGCACCGAGGAGCTGCGGGTGAGCCTCGCCTCCCACCTGCGCAAGCTGCG-TAAG-CGGCTCCTCCGCGATGCCGATGACCTGCAGAAGCGCCTGGCAGTGTACCAGGCCGGGGCCCGCGAGGGCGCCGAGCGCGGCCT

4.4. Morphology

The morphology was investigated by light microscopy 1 or 2 days after thawing one vial from the bank.

5. Karyotyping

For karyotyping, cells were treated with colcimid (Gibco) when 60–80 % confluent, then incubabted with 0.075 M KCl for 30 min at 37 °C and fixed with 1:3 acetic acid:methanol and sent for G-band karyotyping (University of Tübingen). At least 15 metaphases were counted and 6 of them were structurally evaluated by G-banding and a banding quality of 400–500. STR analysis.

For the STR analysis, DNA was extracted (Qiagen) and analyzed using the AmpFLSTR Identifiler PCR Amplification kit (Applied Biosystems).

6. SNP array analysis

A SNP array analysis for the three clones was carried out by Life&-Brain in Bonn, Germany, using a "GSA-Multi-Disease24v2-0" chip.

7. Trilineage differentiation

For trilineage differentiation, iPSCs were split with accutase into a well of a 12-well plate with E8 medium on matrigel in different densities: 200,000 cells/cm^2 for ecto- and endoderm and 50,000 cells/cm^2 for mesoderm. For ectodermal differentiation, the medium was changed to neural induction medium (50 % DMEM F12 and 50 % Neurobasal medium, 1 × B27 without retinoic acid, 1 × N2 supplement, 1 × glutamax, 1 × Pen/Strep (all Gibco), 10 μM SB431542, 0.1 μM LDN193189 (both from Selleckchem)) on day one. The medium was changed every day until day six. For endodermal differentiation, the medium was changed to MCDB131–1 medium (MCDB131 medium, 1.5 g/L NaHCO3, 1 × glutamax, 1 × Pen/Strep (all Gibco), 10 mM glucose (Sigma), 0.5 % BSA) on day one including 3 μM CHIR99021 (Selleckchem) and 100 ng/mL Activin A (Cell Guidance Systems). On day two, CHIR99021 was withdrawn and MCDB131–1 medium with activin A was changed every day until day six. For mesodermal differentiation, the medium was replaced by mesodermal induction medium (APEL medium (Gibco), 25 μg/mL Activin A (Cell Guidance Systems), 30 ng/mL BMP4 (Peprotech), 50 ng/mL VEGF (peprotech), 1.5 μM CHIR99021 (Selleckchem)), which was left on the cells for two days. On day three, the medium was changed to vascular specification medium (APEL medium, 50 ng/mL VEGF, 10 μM SB431542 (Selleckchem)), which was then changed every day until day six.

8. Flow cytometry

Flow cytometry was used for pluripotency and for trilineage differentiation analyses. The cells were prepared in 12 well plates. At the day of the analysis, the cells were treated with 0.5 mL accutase (10 min incubation at 37 °C). The cells were mixed with 1.5 mL 2 % BSA solution and resuspended with a pipet to generate single cells. 200,000 cells were spun down at 120 g. The pellet was resuspended in 0.5 mL of Foxp3 fixation/permeabilization working solution (diluted 1:3, Invitrogen) and incubated at RT for 30 min. Cells were washed in 1 mL 1 × permeabilization buffer, centrifuged at 120g and resuspended in permeabilization buffer containing the antibodies in the respective concentration (Table 2). After 45 min incubation at room temperature, the cells were centrifuged and resuspended in 200 μL permeabilization buffer. The cells were transferred to one well of a 96 well plate with round bottom shape and washed three times by repeating the centrifugation and resuspending step. The analysis was run at a calibrated flow cytometer (analyze 50.000 cells at high speed in 150 μL buffer).

9. Bionano Optical Genome Mapping (OGM)

Briefly, OGM was performed on ultra-high molecular weight (UHMW) genomic DNA (gDNA) labeled with sequence-specific labelling using Direct Label Enzyme (e.g., DLE-1) as previously reported (Iqbal et al., 2023). The labelled DNA was then loaded on the Bionano Genomics Saphyr System to collect 100x coverage. Next, digitalized single-molecule images were assembled de novo into genome maps using the assembly pipeline developed by the Bionano Solve pipeline (v3.7) with default settings. All structural variations (SVs), including deletions, insertions, inversions, and translocations, were detected compared to the hg38 reference by the Variant Annotation Pipeline of Bionano Solve (v1.7.2). The SVs of interested regions were shown after then being compared to the parental iPSC line BIONi010 by using the dual analysis pipeline.

10. STR analysis

STR analysis table supplied as supplementary material.

11. Cell lines identity testing

Line identity tested using STR analysis and integrity further tested using SNP array analysis.

12. Abnormal karyotype

Normal Karyotype.

CRediT authorship contribution statement

Mansour Haidar: Writing – review & editing, Writing – original draft, Investigation. **Benjamin Schmid:** Methodology. **Agustín Ruiz:** Writing – review & editing, Investigation. **Andreas Ebneth:** Supervision. **Alfredo Cabrera-Socorro:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scr.2024.103414.

References

Arboleda-Velasquez, J.F., Lopera, F., O'Hare, M., Delgado-Tirado, S., Marino, C., Chmielewska, N., et al., 2019. Resistance to autosomal dominant Alzheimer's disease in an APOE3 Christchurch homozygote: a case report. Nat. Med. 25 (11), 1680–1683.

Iqbal, M.A., Broeckel, U., Levy, B., Skinner, S., Sahajpal, N.S., Rodriguez, V., Stence, A., Awayda, K., Scharer, G., Skinner, C., Stevenson, R., Bossler, A., Nagy, P.L., Kolhe, R., 2023 Mar. Multisite Assessment of Optical Genome Mapping for Analysis of Structural Variants in Constitutional Postnatal Cases. J. Mol. Diagn. 25 (3), 175–188.

https://doi.org/10.1016/j.jmoldx.2022.12.005. PMID: 36828597; PMCID: PMC10851778.
Rajabli, F., Beecham, G.W., Hendrie, H.C., Baiyewu, O., Ogunniyi, A., Gao, S., et al., 2022. A locus at 19q13.31 significantly reduces the ApoE epsilon4 risk for Alzheimer's Disease in African Ancestry. PLoS Genet. 18 (7) e1009977.
Schmid, B., Prehn, K.R., Nimsanor, N., Garcia, B.I.A., Poulsen, U., Jorring, I., et al., 2019. Generation of a set of isogenic, gene-edited iPSC lines homozygous for all main APOE variants and an APOE knock-out line. Stem Cell Res. 34, 101349.
Serrano-Pozo, A., Das, S., Hyman, B.T., 2021. APOE and Alzheimer's disease: advances in genetics, pathophysiology, and therapeutic approaches. Lancet Neurol. 20 (1), 68–80.